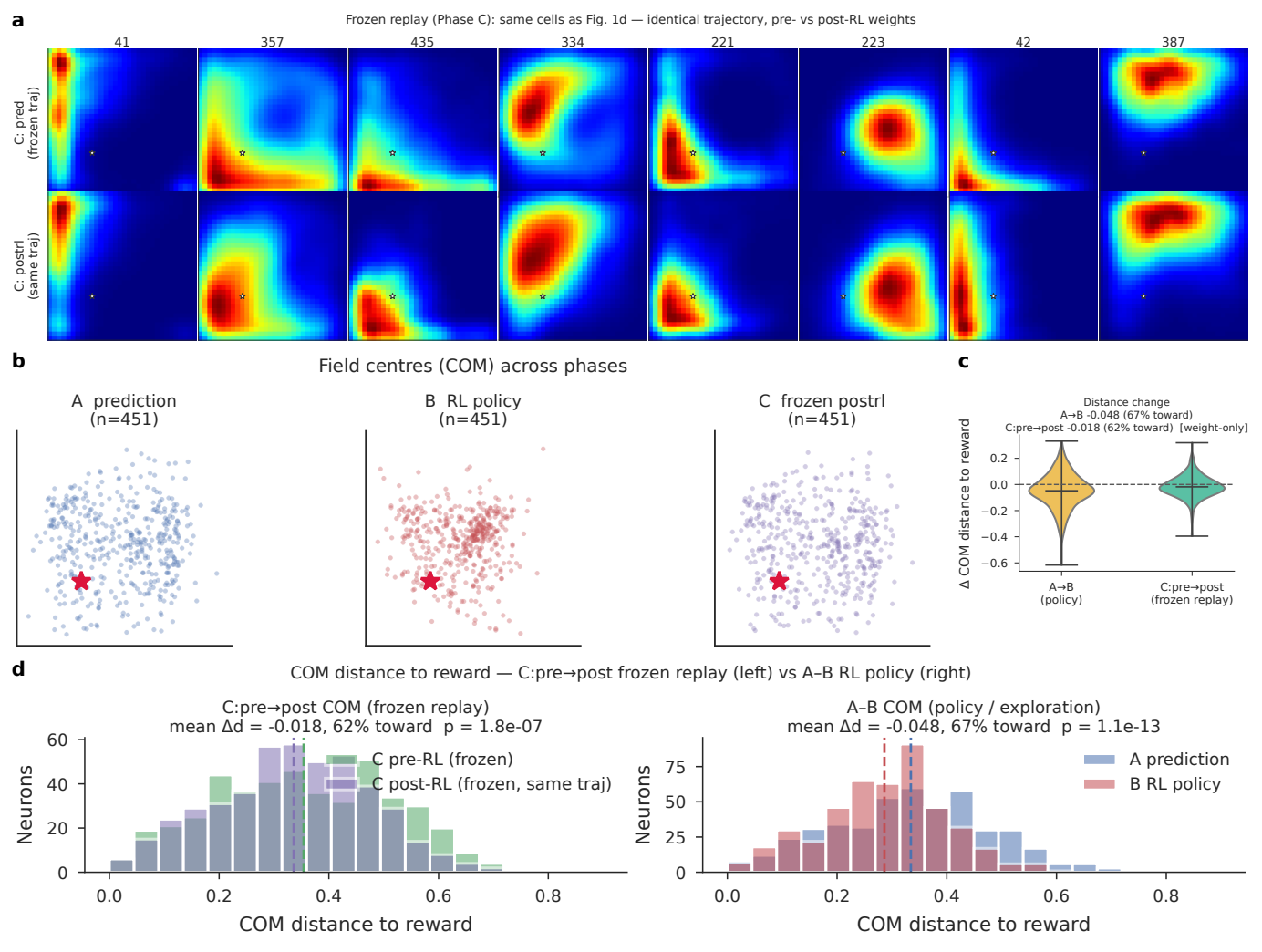

a
Frozen replay (Phase C): same cells as Fig. 1d — identical trajectory, pre- vs post-RL weights
41
357
435
334
221
223
42
387
C: pred (frozen traj)
C: postrl (same traj)
b
Field centres (COM) across phases
A prediction (n=451)
B RL policy (n=451)
C frozen postrl (n=451)
c
Distance change
A→B -0.048 (67% toward)
C:pre→post -0.018 (62% toward) [weight-only]
Δ COM distance to reward
0.2
0.0
−0.2
−0.4
−0.6
A→B (policy)
C:pre→post (frozen replay)
d
COM distance to reward — C:pre→post frozen replay (left) vs A–B RL policy (right)
C:pre→post COM (frozen replay)
mean Δd = -0.018, 62% toward p = 1.8e-07
C pre-RL (frozen)
C post-RL (frozen, same traj)
Neurons
60
40
20
0
0.0
0.2
0.4
0.6
0.8
COM distance to reward
A–B COM (policy / exploration)
mean Δd = -0.048, 67% toward p = 1.1e-13
A prediction
B RL policy
Neurons
75
50
25
0
0.0
0.2
0.4
0.6
0.8
COM distance to reward